ALEXIS WEBER

Bioinformatics Developer | Assay Design & Genomics Workflows | Python and Cloud

LinkedIn: https://www.linkedin.com/in/alexiswebercompbio/

Developer Page: https://alexisweber.github.io/

PROFESSIONAL SUMMARY

Bioinformatics developer with 7+ years of experience designing genomic analysis pipelines and computational tools bridging experimental biology and software engineering. Extensive background in DNA/RNA-based technology analysis, reference genome modeling and performance evaluation of molecular measurements. Proven ability to develop scalable, test-driven Python and R based methods, collaborate with experimental R&D teams and translate biological requirements into robust analytical solutions.

TECHNICAL SKILLS

Computation and Software Skills:

- Programming and Scripting: Python, R, Bash, Linux/Unix
- Version Control: Git, GitHub
- Front-End: HTML, CSS

Data and Infrastructure:

- AWS Cloud Computing
- Docker, Apptainer, Slurm
- Quarto, Markdown, Shiny apps

EDUCATION

Ph.D., Human Genetics | University of California, Los Angeles

B.Sc., Genomics and Molecular Genetics | Michigan State University

RESEARCH AND DEVELOPMENT EXPERIENCE

Bioinformatics Developer and Ph.D. Researcher | University of California, Los Angeles

- Generated and analyzed the largest, first-ever published trisomy 21 neurodevelopmental single-cell multi-omic (RNA+DNA) atlas of neocortex (~114k cells) and patient-derived primary cells (~45k cells).
- Designed and implemented scalable, reproducible bioinformatics pipelines in Python and R for integrated multi-modal analysis, regulatory inference, and cellular and molecular mechanistic discovery.
- Integrated sequencing data with reference genomes and transcriptome models.
- Developed statistical frameworks to evaluate data quality, assay performance, and copy number variation, improve biological signal specificity, and account for technical variation.
- Established standardized data metrics and interactive dashboards to communicate results across multidisciplinary scientific and non-scientific audiences.
- Worked alongside and in collaboration with wet-lab teams to align experimental design with computational analysis and interpretation.

Bioinformatics Researcher | Van Andel Research Institute

- Developed Python, R, and Linux-based pipelines to analyze differential DNA methylation, chromatin modifier binding and chromatin architecture underlying Alzheimer’s disease neuropathology

- Integrated heterogeneous genomic measurements using reference genome frameworks to identify regulatory disruptions in neurodegenerative diseases.
- Designed computational analyses to support experimental assay interpretation and optimization.
- Collaborated with molecular biologists to iteratively refine computational outputs based on assay and sample behavior.

Undergraduate Researcher | Michigan State University

- Developed Python and Linux-based analysis pipelines to identify differential patterns of RNA editing and structure in *Trypanosoma brucei.*
- Applied statistical modeling and custom scripting to quantify RNA processing events.
- Generated publication-quality visualizations to support biological interpretation.

PUBLICATIONS

Weber, A.†, and Vuong, C.K.†, et al. "A single-cell multi-omic analysis identifies molecular and gene-regulatory mechanisms dysregulated in the developing Down syndrome neocortex". *bioRxiv*. (2025). *Under review at Science*. †Contributed equally.

CERTIFICATIONS

- Fundamentals of AI/ML in Precision Medicine, Stanford University
- Fundamentals of Data Science and Cloud Computing in Precision Medicine, Stanford University
- Biosciences Leadership Training Program Certificate, UCLA

AWARDS

- Graduate Division Third Place Poster Award, Neurology Science Day, UCLA
- NIH Biomedical Big Data T32 Training Grant Award, UCLA
- Undergraduate Research and Arts Forum First Place Award, Michigan State University

MENTORSHIP AND TEACHING

- Organizer and Instructor, Introduction to Python for Scientists Workshop, Van Andel Research Institute
- Research Mentor, Amgen Scholars Award Program
- Teaching Assistant, Computational Biology and Genomics, UCLA